Verigate PortalDot

Authorization middleware for AI-agent payments on PortalDot V3

A proposal is not authority.

Verigate evaluates the action. The bridge verifies signed evidence. A dedicated Solidity guard enforces the exact approved transfer.

VERIGATE	BRIDGE	PORTALDOT
Ed25519 decision	Exact action binding	EIP-712 guard

TESTNET

420420777

PortalDot V3 EVM testnet

Contract deployed • Receipt verified

Track fit

Modular Cloud Solutions and Middleware

Built around a narrow, reusable control point between AI intent and token execution.

Preventing unrestricted agent spending

The authorization must bind the decision to the exact thing the executor is allowed to do.

01 RECIPIENT SUBSTITUTION

An approved payment must not be reused with a different destination.

02 AMOUNT TAMPERING

The amount must remain the same from signed intent to contract call.

03 REPLAY OR EXPIRY

A valid authorization should not be usable again or after its deadline.

Design principle

A signed ALLOW is necessary but not enough: its authority must be short-lived, narrowly scoped, and bound to a recipient, amount, chain and contract.

A narrow path from decision to execution

Off-chain verification and on-chain enforcement have separate, explicit trust boundaries.

1

AGENT INTENT

Proposed action

2

VERIGATE

Signed ALLOW + execution authorization

3

BRIDGE

Verify Ed25519; bind exact action

4

EIP-712

Chain, guard, recipient, amount, nonce, deadline

5

GUARD

Reject invalid/replayed or altered transfer

WARN / BLOCK / invalid signature / mismatched action

The bridge refuses to sign; the contract cannot independently override this off-chain decision.

Why PortalDot V3 — and where this project fits

The project uses PortalDot's execution environment; it does not overstate a cross-chain bridge that is not implemented.

- **EVM-compatible contract execution**

  The guard is written in Solidity and uses EIP-712 signatures bound to chain ID and contract address.

- **Substrate-aware operational checks**

  The tooling separately checks the configured Substrate WebSocket endpoint and genesis hash before sensitive network operations.

- **Reusable middleware contribution**

  Other agent-enabled applications can reuse a strict authorization boundary instead of giving an agent a direct, unrestricted transfer path.

TRACK

Modular Cloud Solutions and Middleware

Not claiming:

A deployed multi-chain asset bridge, verified wallet mapping, production readiness, or an independent audit.

V2 execution evidence on PortalDot V3

The recorded positive path is real; limits and wallet integration are not overclaimed.

NETWORK

PortalDot V3 EVM testnet

CHAIN ID

420420777

CURRENT CONTRACT

PortalDotAgentGuardV2

CONTRACT ADDRESS

0x87C494a5...10E046C6

DEPLOYMENT TX (RECORDED)

0x331c1fc0...3f5d3494b

TRANSFER TX

0xaf3df785...f2e90f5551e

TRANSFER RESULT

Success • block 149241

READ-BACK

V2 policy

Recipient allowlist

0.1 POT / transfer

0.5 POT per resettable 3600-second interval.

TRUST BOUNDARY

Ed25519 stays off-chain; V2 checks the bridge EIP-712 signature.

Evidence and unresolved limits

The V2 positive execution path is recorded; negative policy cases are local tests.

LOCAL

contract tests

ALLOW

signed artifacts accepted

BLOCK

denied path refused

- **Pre-broadcast checks passed**

  The executor re-simulated the exact signed authorization and refreshed the fee budget before sending.

- **Public V2 transfer recorded**

  Receipt status 1 at block 149241 and one matching TransferExecuted event, queried through the configured RPC.

IMPORTANT DISTINCTION

The transfer was 0.1 POT. Sender balance fell from 416.1965 to 408.3687 POT across an isolated block, exactly matching the 7.8278 POT receipt-derived amount. One configured RPC only; recipient balance increased from 0 to 0.1 POT across the isolated block. Runtime execution-fee vs refundable storage-deposit components were not separately determined.

What remains before wider adoption

The next work is evidence and user validation, not another unproven feature.

- **Investigate execution cost**

  The sender debit was reconciled as 7.8278 POT for a 0.1 POT transfer, about 78× the transfer value. The reason for this cost and the runtime execution-fee/storage-deposit split remain unresolved; do not claim cost-effective operation.

- **Prove negative policy paths**

  The cap and allowlist rejection cases are backed by local Solidity tests, not recorded public-testnet reverts.

- **Validate ecosystem need**

  No PortalDot Wallet/DaaS integration, team endorsement, or external user adoption is currently confirmed.

ECOSYSTEM VALUE

A reference middleware path from Verigate Ed25519 authorization off-chain to EIP-712 enforcement by a PortalDot V3 guard. This is not an integration into PortalDot Wallet or DaaS.

LIMITATIONS

Testnet prototype, no audit. One configured RPC; recipient balance delta confirmed on that RPC. Runtime cost breakdown remains unknown. The resettable cap is not a sliding window. Demo video and user validation remain pending.